DCSO

BSI Schulung: MISP in Practice

One-Day Training: From the Basics to Running Your Own Instance

Alexander Heidorn

2026

TLP:CLEAR

Workshop Logistics

Making the Day Work for Everyone



Breaks

A 60-minute lunch break and two shorter breaks during the day.



Questions

Time for questions after every chapter. Otherwise, just raise your hand.



Materials

The slides will be shared after the workshop.



Photos

Allowed for personal use. Note: content may be TLP-restricted.

References: MISP Book (circl.lu/doc/misp) and MISP Cheat Sheet (misp-project.org/misp-training/cheatsheet.pdf)

Agenda (1/2)

1

What Is MISP?

Introduction to the MISP Basics

2

Sharing with MISP

Concepts, Visibility, Technology and Communities

3

Deep Dive: Working with MISP

Warninglists, Blocklists, Feeds, Galaxies and Tag Collections

4

MISP Use Cases

From Use Case to Detection

Agenda (2/2)

5

Connecting MISP

MISP API & PyMISP

6

Operating MISP

Options for Running Your Own Instance

7

Stretch Topics

Reports, Templates, Decaying, Dashboards, Modules, Workflows, ZeroMQ, AI

Schedule

Suggested Timetable for the Training Day

Time	Block	Content
09:00	Start	Welcome, trainer introduction, logistics
09:00 – 13:00	Morning	1 What Is MISP? · 2 Sharing with MISP · 3 Deep Dive · short break
13:00 – 14:00	Lunch Break	
14:00 – 17:00	Afternoon	4 Use Cases and Detection · 5 Connecting · 6 Operating · short break
17:00	Wrap-Up and End	Key takeaways, questions, feedback

Chapter 7 (Stretch Topics) is optional and used only if time allows.

1 What Is MISP?

Introduction to the MISP Basics



MISP at a Glance

Open Source TIP

A platform for storing, correlating and sharing indicators of compromise and threat intelligence, e.g. on attackers, campaigns or financial fraud.

ORGANISATIONS
in the CIRCL communities alone

2,000+

LICENCE
Free open source software

AGPL

DEVELOPMENT
Together with the community

CIRCL

TLP:CLEAR

Core Functions



Collect & Structure

Capture indicators, context and analyses in one consistent data model.



Correlate

Automatically detect overlaps between events, feeds and attributes.



Share

Exchange information with partners and communities in a controlled way.



Use

Push indicators straight into security systems via API and exports.

Data Model: Events, Objects and Attributes

A Structure that Carries Context

EVENT Event Info · Date · Threat Level · Analysis · Distribution · Tags

Object: file

- filename: invoice.pdf.exe
- sha256: 9f86d081884c7d65...
- size-in-bytes: 48213

Object: domain-ip

- domain: invoice-check.example
- ip: 203.0.113.10
- last-seen: 2026-10-14

STANDALONE ATTRIBUTES

- url: https://invoice-check.example/login
- email-src: billing@invoice-check.example

Objects can be linked via object references, e.g. file "drops" / domain-ip "communicates-with".

Event

Container for an incident, a campaign or a report, including metadata.

Object

Template that groups related attributes, e.g. file, domain-ip or email.

Attribute

A single value with category, type, IDS flag, comment and its own distribution.

Context

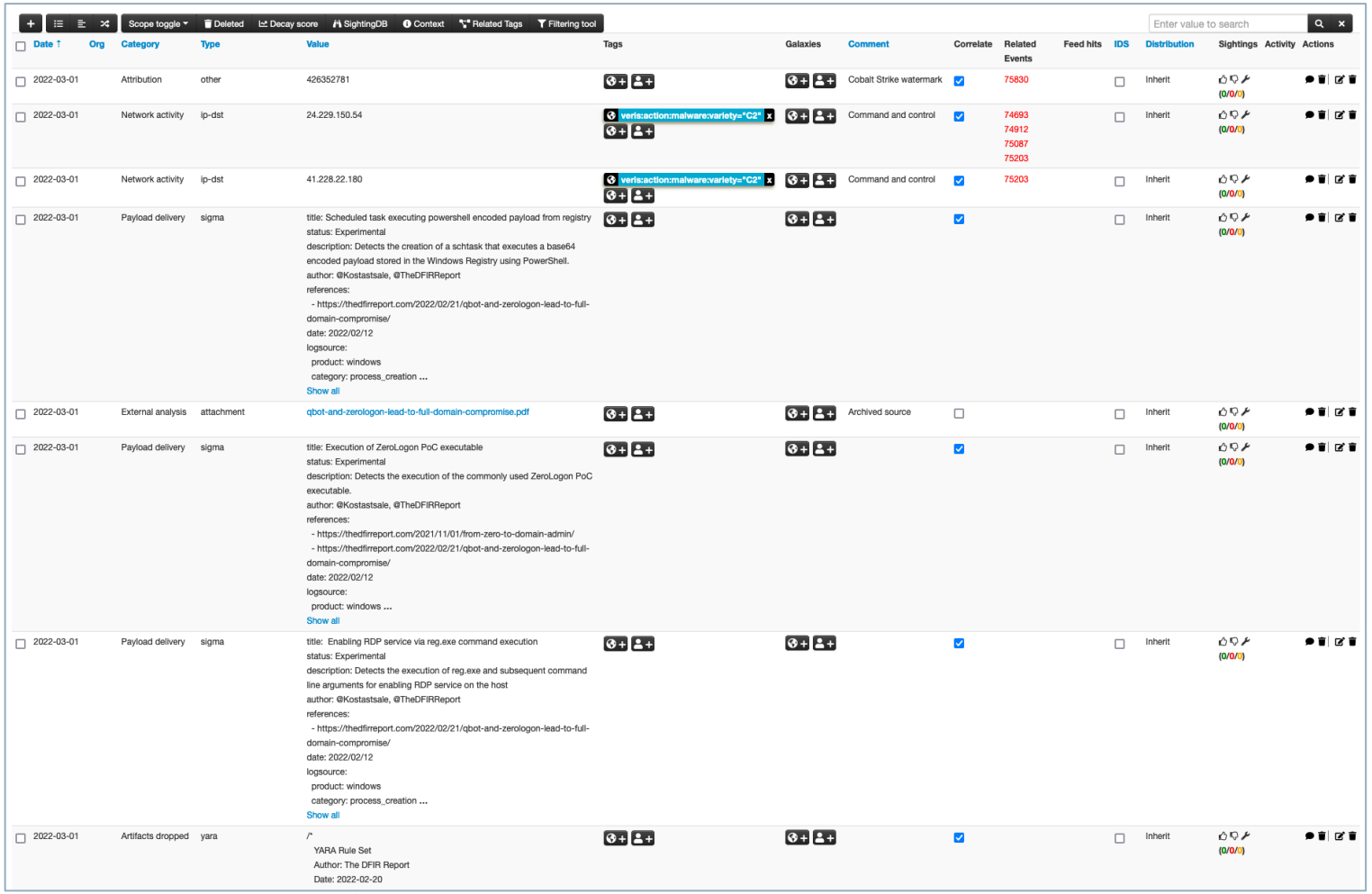
Tags, galaxies, event reports and sightings enrich every level.

What an Event Looks Like in MISP

Attributes, Tags, Correlations and IDS Flag in One View

For each value, the attribute list shows:

- Category and type
- Tags and galaxies at attribute level
- Correlations (related events) and feed hits
- IDS flag and distribution
- Sightings as feedback from usage



Anatomy of an Attribute

IDS Flag: Yes or No?

The IDS flag (to_ids) decides whether an attribute ends up in detection exports. Before setting it, ask:

- Is the value suitable for automated detection?
- Or only for manual hunting?
- Or does it merely provide context?

Rule of thumb: Only set IDS for actionable indicators; otherwise expect high false-positive rates, e.g. with shared-hosting IPs.

Field	Meaning	Example
Category	Functional classification	Network activity
Type	Technical type, drives validation & export	ip-dst
Value	The actual value	203.0.113.10
IDS (to_ids)	Suitable for automated detection	true
Distribution	Visibility, at most that of the event	Inherit event
Comment	Context for analysts	C2 of the phishing wave
First / Last seen	Observation period	2026-10-01 – 2026-10-14
Correlate	Correlation on or off	on

Object Templates and References

Reusable Structures Instead of Loose Values

TEMPLATE **domain-ip**

KEY · TYPE · RULE

domain	domain	required one of
ip	ip-dst	required one of, multiple
port	port	multiple
first-seen	datetime	optional
text	text	optional

Templates

300+ templates in the misp-objects repository define keys, types and required fields.

Validation

MISP enforces required keys and types, so objects stay comparable across organisations.

Object References

Relationships such as drops, communicates-with or downloaded-from link objects and attributes.

Custom Templates

Create your own templates locally for internal data and share them with partners.

Taxonomies: Consistent Classification

Machine-Readable Tags Instead of Free Text

Taxonomies are vocabularies of machine tags. They add context, drive workflows and act as filters for sync, export and automation.

STRUCTURE OF A MACHINE TAG



- Only enable the taxonomies you need
- Custom taxonomies as JSON, e.g. dcso-sharing:event-type="Analysis"
- Prefer tagging at event level

Taxonomy	Purpose	Example Tag
TLP	Sharing restrictions	tlp:amber
FIRST IEP	Handling, actions, licence	iep:commercial-use="MUST NOT"
Workflow	Processing status	workflow:state="draft"
Admiralty Scale	Source & information reliability	admiralty-scale:source-reliability="b"
VERIS	Context and incident description	veris:actor:motive="Financial"
OSINT	OSINT classification	osint:source-type="blog-post"

Recommended taxonomies to get started

Galaxies: Context Beyond Tags

How Galaxies Extend What Taxonomies Can Express

	Taxonomy Tag	Galaxy Cluster
What It Is	Label from a controlled vocabulary	Entry in a shared knowledge base
Content	Namespace, predicate, value	Description, synonyms, meta data, references
Relationships	None	Between clusters, e.g. actor uses malware
Main Purpose	Classification, handling, workflow	Context on actors, tools and techniques
Example	tlp:amber	misp-galaxy:threat-actor="APT28"

POPULAR GALAXIES

Threat Actor | **Malware** | **Tool** | **Ransomware** | **MITRE ATT&CK** | **Sector** | **Country**

Same Mechanism

Attached to events and attributes just like a tag.

Much More Context

Synonyms map different vendor names to one actor or malware family.

Connected Knowledge

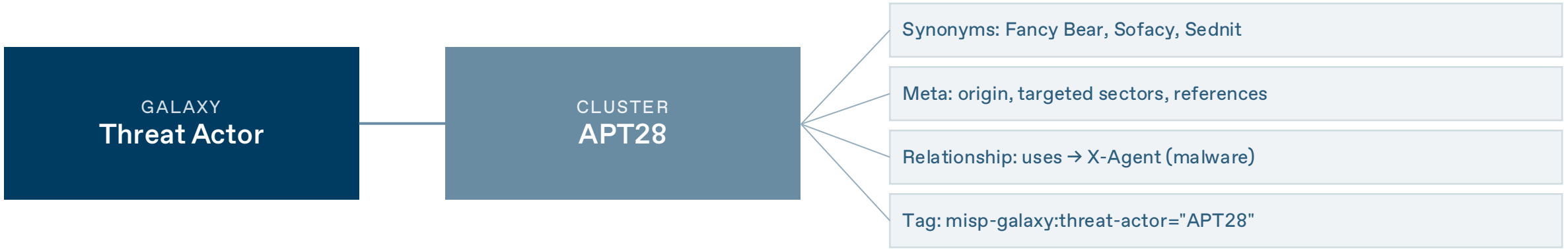
Relationships show which actor uses which tools and techniques.

Up Next

How a galaxy cluster is built; MITRE ATT&CK follows in chapter 3.

Galaxies: Context on Actors and Tools

A Shared Language for Threat Intelligence



Many Galaxies

Threat actor, malware, tool, ransomware, sector, country, MITRE ATT&CK and many more.

Compared to Taxonomies

Galaxies provide context and relationships, not just a label.

Extensible

Create your own galaxies and clusters locally and share them with communities.

Taxonomy or Galaxy? When to Use Which

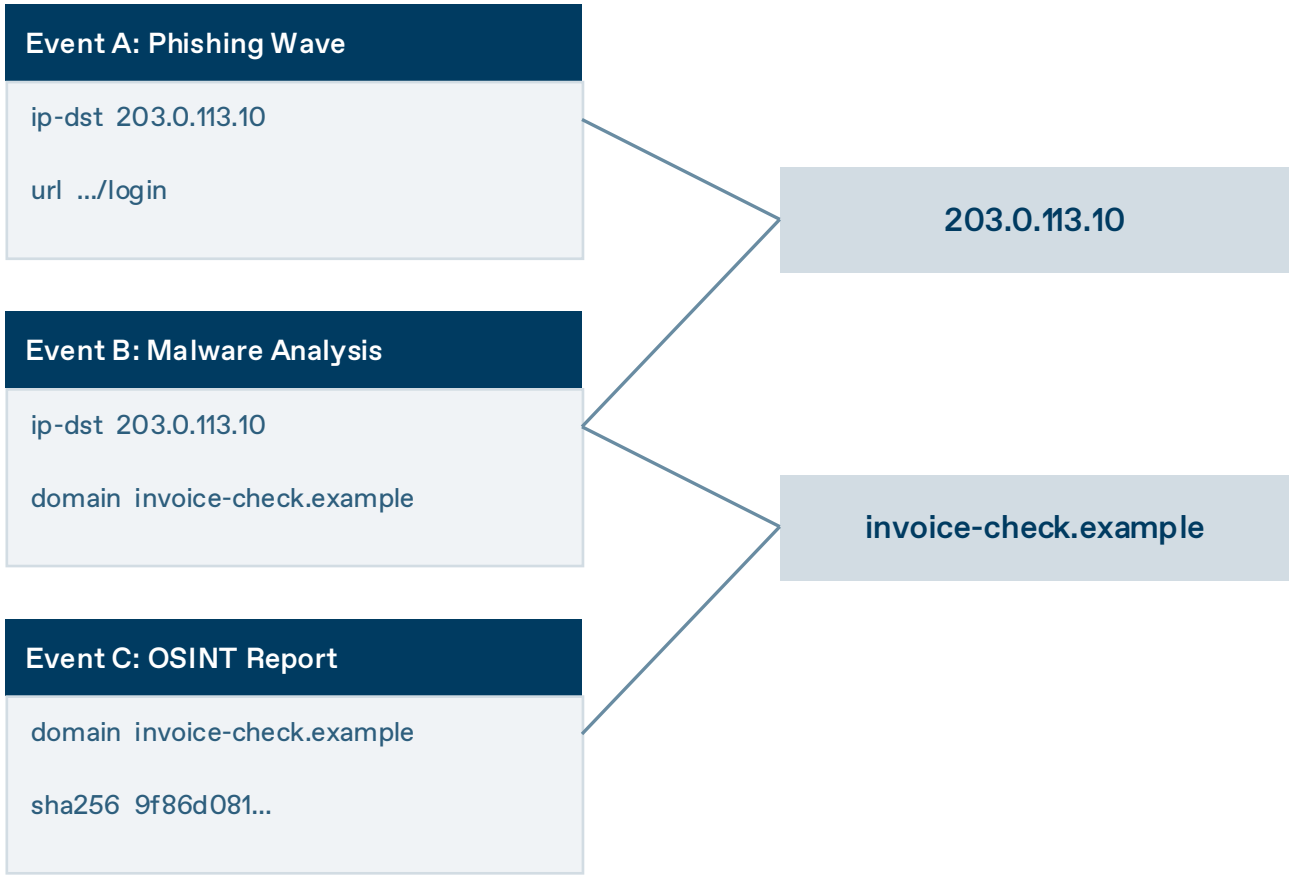
Handling Instructions vs. Knowledge About the Threat

Taxonomy How should this be handled?		**Galaxy** What is this about?	
May I share this with partners?	tlp:amber	Which actor is behind it?	threat-actor="APT28"
How reliable is the source?	admiralty-scale:source-reliability="b"	Which malware family is involved?	malpedia="Emotet"
What is the processing status?	workflow:state="draft"	Which technique was used?	mitre-attack-pattern="Phishing - T1566"

Rule of thumb: Taxonomies tell you how to handle information, galaxies tell you what it is about. Most events need both.

Correlations: Spot Connections Automatically

Shared Values Link Events



Automatic

MISP compares values across events, feeds and server caches.

Context, Not Just a Value

Correlations reveal campaigns, shared infrastructure and tooling.

Avoid Bloat

Disable correlation for generic values such as image.jpg, calc.exe or registrar and CRL systems.

With Warninglists & Feeds

Hits on Tor exit nodes or sinkholes provide instant context.

Let's Take a Look in MISP

Live Demo: What We Just Covered, in Reality



WHAT WE WILL LOOK AT

1. Event index: find and filter events
2. An event: attributes with category, type and IDS flag
3. Objects and their references
4. Taxonomy tags and galaxy clusters on the event
5. Correlations: related events and shared values

LIVE DEMO: APPROX. 10 MINUTES



Event ID vs. Event UUID

Local Number or Global Identity: Use the Right One

Event ID

Local number, e.g. 1234

- Auto-incremented by each instance
- Unique only on this instance
- Same event has a different ID elsewhere
- Fine for quick local lookups and URLs

Event UUID

Global identity, e.g. 5f1c…a83e

- Assigned once when the event is created
- Identical on every instance, feed and export
- Attributes, objects and reports have UUIDs too
- Use for partners, scripts, blocklists, dedup

SAME EVENT ON TWO INSTANCES

Instance A

ID 1234 **UUID 5f1c…a83e**

Instance B

ID 87 **UUID 5f1c…a83e**

Different ID, identical UUID

Rule of thumb: The ID is for you, the UUID is for everyone else. Reference events by UUID when sharing, scripting or blocking.

Event Lifecycle: From Draft to Shared Information

Clear Responsibilities in Every Phase



Create

Set event info, date, threat level, analysis status and distribution.

Enrich

Add attributes and objects, e.g. via freetext import or an enrichment module.

Review & Tag

Clean up warninglist hits, check IDS flags, assign taxonomies and galaxies.

Publish

Release the event: from now on it is synced, exported and notified.

Share

Distribution to connected instances according to distribution and sharing groups.

Maintain

Process sightings and proposals, clean up outdated indicators.

Important: Only "Publish" synchronises an event and triggers notifications. Changes after publishing require publishing again.

Publishing: Sharing Events with Others

What Happens When You Click Publish

What Publish Triggers

1. **Push** Sent to connected instances per distribution and push filters
2. **Pull & Exports** Partners and restSearch with published=1 pick it up on the next run
3. **E-Mail Alerts** Notification to subscribed users, optionally PGP-encrypted
4. **ZeroMQ / Kafka** Publish message for real-time consumers
5. **Workflows** Workflows on the event-publish trigger run, and may block

Before You Publish

- [ ] Distribution and sharing group set deliberately
- [ ] TLP and other handling tags present
- [ ] IDS flags reviewed, warninglist hits cleaned up
- [ ] No personal or internal data left in the event
- [ ] Event info, tags and context complete

Good to know: "Publish (no email)" skips alerts. Changes after publishing reach partners only when you publish again.

Getting Data into MISP

Several Ways to Create and Find Events

Freetext Import

Paste text, MISP detects types and checks warninglists.

Objects & Templates

Structured forms for files, e-mails, domains and more.

Event Templates

New 2026: pre-filled events for recurring cases.

Import Modules

E-mail, CSV, OpenIOC, STIX and more via misp-modules.

Feeds & Sync

Events arrive automatically from partners and sources.

API & PyMISP

Scripts and tools create events programmatically.

Finding Data

- Event index: filter by org, tags, date, published
- Search attributes across all events
- Event view: filter attributes, show context
- Event graph and timeline for relationships

Recommended Event Types

Consumers See at Once What an Event Is Meant For

Event Type	Description	Use For	Don't Use For
Observation	Indicators on one entity, e.g. an e-mail campaign	Timely distribution, automated blocking	Enrichment, context
Incident	First-hand account of a security incident	Context, enrichment, hunting	Automated blocking
Report	Preservation of an original report incl. context	Traceability, enrichment	Unreviewed automation
Analysis	Enrichment of other events by analysts	Context, enrichment	Automated blocking
Collection	Set of mostly unrelated IoCs	Periodic distribution to systems	Correlation, enrichment

The event type can be expressed as a tag via a custom taxonomy, e.g. dcso-sharing:event-type="Observation".

Best Practices

1

Get Oriented First, Then Contribute

Learn the dashboard, event index and filters first.

2

Choose Distribution Deliberately

Start restrictive and widen only after review.

3

Context over Volume

Event info, comments and tags make indicators usable.

4

Set the IDS Flag with Care

Only mark reviewed, detection-ready values.

5

Don't Publish Too Early

Get a review first; publishing triggers sync and e-mails.

Key Takeaways · Chapter 1

What Is MISP?



WHAT WE HAVE LEARNED

- [ ] Explain how events, objects and attributes fit together
- [ ] Decide when an attribute deserves the IDS flag
- [ ] Classify events with taxonomies and add context with galaxies and their clusters
- [ ] Use correlations to spot shared infrastructure across events
- [ ] Take an event from draft to publish, knowing that publishing triggers sync

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



2 Sharing with MISP

Concepts, Visibility and Technology

Two Levels of Sharing

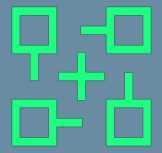
What Users Decide and What MISP Does with It Technically



Level 1: User View

When creating an event in the UI

- Choose the distribution level and sharing groups
- Understand who sees an event
- Do's and don'ts, delegation
- Build active communities



Level 2: Technical View

How MISP distributes events between instances

- Synchronisation via PUSH and PULL
- Filters and filter ownership
- Internal vs. external and downgrading
- Connection setup and sharing setups
- Cerebrate and misp-guard

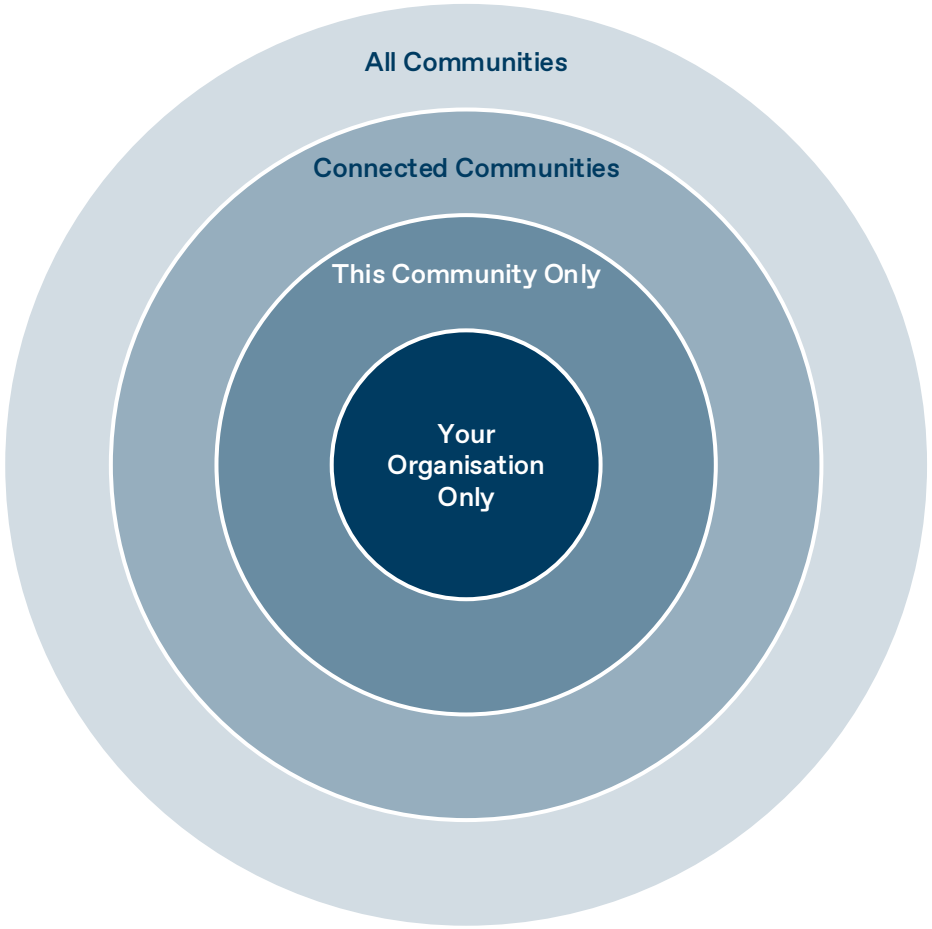
Audience: Level 1 concerns everyone who creates events. Level 2 is mainly for admins and those responsible for sync connections.

2 Level 1: User View

Sharing When Creating Events

Distribution Levels: Who Sees What?

From the Inside Out: The Wider the Ring, the More Recipients



Stays on Your Own Instance

Your Organisation Only
Core: only users of your own organisation see the event.

This Community Only
Inner ring: all organisations on this instance.

Shared with Other Instances

Connected Communities
Additionally, organisations on directly connected instances.

All Communities
All connected communities, with no limit on reach.

Also: Sharing group (explicit list of recipients) and inherit event (attributes and objects take over the event's distribution).

Sharing Groups: Share Precisely

Define Exactly Which Organisations Receive an Event

Event	Sharing Group	Org A	Org B	Org C
Event 1	Sharing Group 1	✓	✓	–
Event 2	Sharing Group 2	✓	✓	✓
Event 3	Sharing Group 3	✓	–	✓
Event 4	Sharing Group 4	–	–	✓

Example: which organisation sees which event?

Benefits

- Exact choice of recipients
- Can be distributed to other instances and extended there

Watch Out For

- Complexity grows with the number of groups and instances
- Member organisation names become visible to other instances
- Review groups regularly to keep them current

Who Sees an Event?

Strictest Rule Wins

Whether a user sees an event or attribute depends on several factors that apply together.

EXAMPLE

Event: **Connected Communities**

Attribute: **Your Organisation Only**

Result: the event is shared, the attribute stays within your own organisation.

Distribution

Event, object and attribute level. The most restrictive level applies.

Organisation & Role

The creating organisation always sees its data; site admins see everything on the instance.

Sharing Group

Only listed organisations and instances receive the data.

Publish Status

An event only leaves your instance once published; drafts stay local.

Attribute Distribution

Individual attributes or objects can be shared more restrictively than the event.

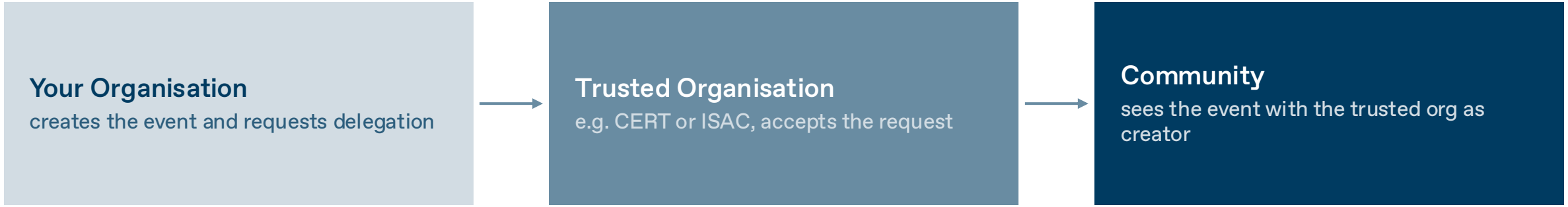
Do's and Don'ts When Sharing

Trust Is Built on Quality

Do's	Don'ts
Set and respect TLP and IEP consistently	Share unreviewed bulk data without context
Provide context: source, time frame, confidence	Pass on personal or customer data carelessly
Check indicators against warninglists before sharing	Leak internal taxonomies and tags unintentionally
Return sightings and feedback to the creator	Redistribute licensed commercial feeds
Correct events and keep them up to date	Set the IDS flag on generic values

Delegation: Share Without Attribution

Let a Trusted Organisation Publish on Your Behalf



When to Use It

Incidents you want to share without revealing that you were affected.

Who Decides

The receiving organisation sets the final distribution and publishes.

Keep in mind: Remove internal context such as hostnames or customer names before requesting delegation. The content itself still reveals what you saw.

Working in Communities

See More Together than Alone

Sector Communities (ISACs)

Sector-Specific

For Companies in the Same Sector



- Focus: Sector Campaigns
- Model: Sharing Groups

Detect threats against your own sector early and assess them together.

CERT & CSIRT Networks

Response Teams

For Incident Responders and CERTs



- Focus: Acute Incidents
- Model: Sync & Feeds

Fast exchange on current incidents through established trust relationships.

Providers & Trust Circles

Curated Intelligence

For Customers and Partner Networks



- Focus: Qualified IoCs
- Model: Pull with Filters

Quality-assured intelligence with clear rules on use and redistribution.

Building an Active Community

From Passive Consumers to Contributors

Activating "Leechers"

- Feedback counts: sightings, proposals and opinions are contributions
- Start small: share what you already see, e.g. phishing
- Make contributors visible and recognise them
- Use delegation where attribution is a blocker

Dispelling Common Myths

"We have nothing to share"
Your own observations are often the most valuable data.

"Our data is not good enough"
Context and feedback improve data together.

"Legal won't allow it"
GDPR explicitly supports sharing for security (recital 49).

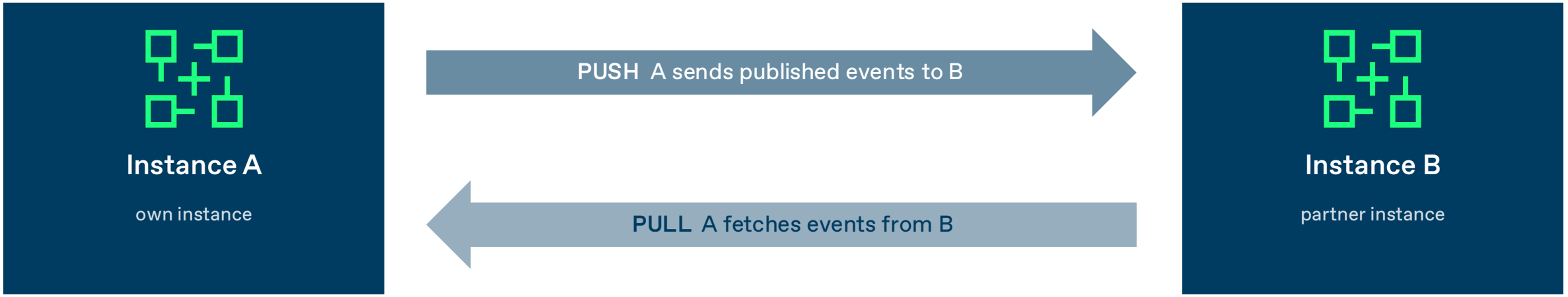
Ground rules: Agree on vocabularies (taxonomies, galaxies), quality expectations and how to handle commercial or personal data before the community grows.

2 Level 2: Technical View

How MISP Distributes Events

Synchronisation: PUSH and PULL

Two Ways to Exchange Events Between Instances



PUSH

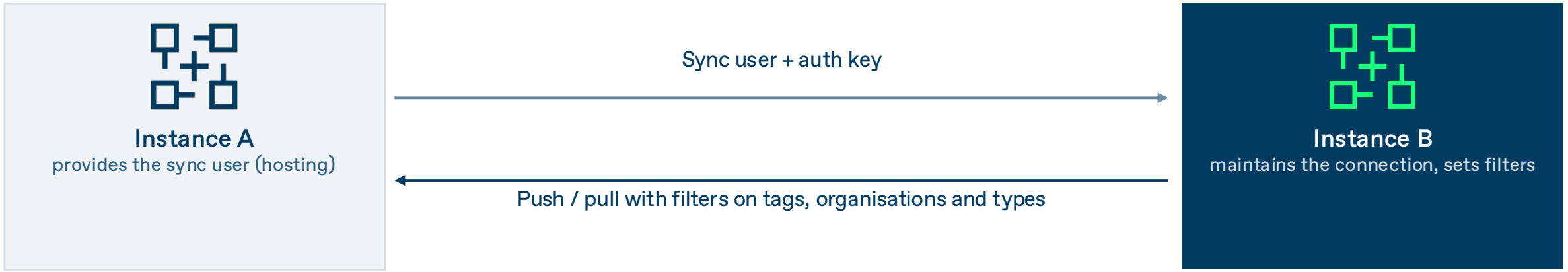
- Target must be reachable from A (firewall, proxy, port 443)
- Usually runs every time an event is published
- Requires a sync user and auth key on instance B
- A can set filters for outgoing events

PULL

- Target must be reachable from A (firewall, proxy, port 443)
- Usually runs periodically, e.g. via scheduled task or cron
- Requires a sync user and auth key on instance B
- A can set filters for incoming events

Filters and Filter Ownership

Whoever Maintains the Connection Controls the Filters



Hosting Instance (A)

Cannot filter what the sync user fetches. It sees everything per distribution and sharing groups.

Connected Instance (B)

Sets push and pull filters as needed.

In practice: Filters such as "not tlp:amber" or "only org X" prevent accidental sharing. For each connection, check who controls which side.

Internal vs. External and Event Downgrading

How Distribution Changes Across Multiple Hops



Distribution at A	A → B (internal)	B → C (external)	C → D (external)
Your Organisation Only	✓ unchanged	✗ not shared	–
This Community Only	✓ unchanged	✗ not shared	–
Connected Communities	✓ unchanged	✓ becomes "This Community"	✗ not shared
All Communities	✓ unchanged	✓ unchanged	✓ unchanged

Remember: Only "Connected Communities" is downgraded. "Internal instance" shares everything unless filtered and is suited to mirroring your own instances.

Connection Setup in Five Steps

From Sync User to First Pull

1. **Request a Sync User**
   The partner creates a user with the "Sync user" role on their instance and sends the auth key.

2. **Create the Server Connection**
   Sync Actions › List Servers › Add Server: URL, auth key and remote organisation (UUID).

3. **Choose the Connection Type**
   Set push, pull, internal, caching and, if needed, certificate or self-signed.

4. **Define Filters**
   Set push and pull rules on tags, organisations and types.

5. **Test & Activate**
   Run Test Connection, start the first pull and check the logs.

Common Pitfalls

- Firewall or proxy blocks port 443
- TLS certificate not trusted
- Auth key belongs to the wrong user or role
- Remote organisation with the wrong UUID
- Overly broad pull rules flood the instance
- Scheduled tasks do not run reliably

Sharing Setups Compared

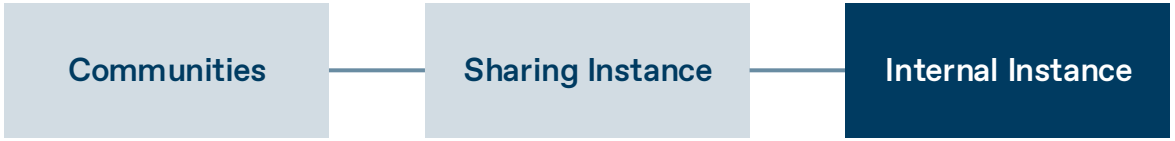
Keep Internal Data Safely Apart from Shared Content

Setup 1: Direct Connection



+ Quick and easy to implement
– Higher risk of sharing internal events or commercial feeds
– PUSH requires a sync account on the internal instance

Setup 2: Dedicated Sharing Instance



+ Clear separation of internal and shareable data
+ Additional filter layer, e.g. via tag filters
– Higher maintenance effort for a second instance

Let's Take a Look in MISP

Live Demo: Sync Connections and Their Filters



WHAT WE WILL LOOK AT

1. Server overview: existing connections under Sync Actions › List Servers
2. A connection: URL, remote organisation, auth key and sync options
3. Push and pull rules: filters on tags and organisations, e.g. not tlp:red
4. Test Connection: reachability, version and permissions

LIVE DEMO: APPROX. 10 MINUTES



Key Takeaways · Chapter 2

Sharing with MISP



WHAT WE HAVE LEARNED

- [ ] Choose the right distribution level, from your organisation only to all communities
- [ ] Use sharing groups when exactly defined recipients should get an event
- [ ] Explain why the most restrictive setting wins
- [ ] Describe how PUSH and PULL move events between instances
- [ ] Name who controls the filters on a sync connection

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



3 Deep Dive: Working with MISP

Data Quality and Context

Warninglists

Catch False Positives Early

Lists of known, benign values that MISP takes into account for correlation, display and export.

WARNINGLISTS
in the official repository

230+

LIST TYPES
string, substring,
hostname, cidr, regex

5

Typical Entries

Warninglist	Type	Example
Top domains (e.g. Tranco)	hostname	google.com
Private networks (RFC 1918)	cidr	10.0.0.0/8
Public DNS resolvers	cidr	8.8.8.8
Cloud provider ranges	cidr	AWS, Azure, GCP
Custom list: company IPs	cidr	198.51.100.0/24

What a Hit Looks Like



Screenshot: warninglist hits shown on an attribute in MISP

In the UI

Warning on the attribute and in correlations.

In Exports

enforceWarninglist filters hits out of restSearch.

Built-In Warninglists: 231 Lists Out of the Box

What MISP Ships and Which Lists to Enable

What Ships with MISP

Category	Lists	Examples
Scanners & research	92	Shodan, Censys, CIRCL
Cloud, hosting & CDN	59	AWS, Azure, Cloudflare
DNS, mail & web services	31	Public DNS, CRL, URL shorteners
Top domain rankings	25	Tranco, Cisco, Majestic
Reserved IP ranges	11	RFC 1918, multicast
Hashes & known FPs	10	Empty hashes, EICAR
Anonymisation	3	Tor exit nodes, VPN

By type: 161 cidr · 33 string · 31 hostname · 4 substring · 2 regex. Source: misp-warninglists repository, Oct 2026.

Our Recommendations

Always On
Reserved ranges, public DNS, CRL, empty hashes, common IoC false positives.

Add Selectively
Cloud ranges for your providers and one top-domain list (e.g. Tranco 10k).

Context, Not Noise
Scanner, Tor and VPN hits are signals, so do not filter them blindly.

Mind the Size
1M-entry lists cost memory and lookup time. Enable only when needed.

Keep It Current
Update regularly and add custom lists for your own IPs and domains.

Blocklists: Keep Unwanted Content Out

How They Differ from Warninglists and Correlation Exclusions

Mechanism	Applies To	Effect	Typical Use
Warninglist	Attribute values	Warning, filtered out of exports	Known benign values
Event Blocklist	Event UUIDs	Event is not created or pulled again	Deleted or unwanted events
Organisation Blocklist	Organisations (UUID)	Events from this org are rejected	Low-quality sources
Correlation Exclusion	Attribute values	No correlation for the value	Generic values without meaning

Good to know: Deleted events automatically land on the event blocklist and therefore do not come back via sync.

Stay organised: Blocklists help most when the instance is connected to many communities.

Let's Take a Look in MISP

Live Demo: Warninglists and Blocklists in Action



WHAT WE WILL LOOK AT

1. Warninglist overview: which lists are enabled and what they contain
2. A benign value in an event: how MISP shows the warning
3. Export with enforceWarninglist: the hit is filtered out
4. Delete an event: it lands on the event blocklist
5. Organisation blocklist: rejecting a whole source

LIVE DEMO: APPROX. 10 MINUTES



Tagging Done Right

From Context to Findable Data

The ATT&CK techniques we just saw are tags. Tags are what makes MISP data searchable, shareable and usable for automation, but only if everyone tags the same way.



Why Tags Matter

Find and Filter

Search, restSearch and feeds all filter by tags.

Control Sharing

TLP and local tags decide what leaves your instance.

Drive Automation

Exports, workflows and detection pipelines react to tags.

Coming Up

Tag Collections	Tagging in Practice	Rating Sources
Tag consistently	Local or global	Reliability & credibility

Tag Collections: Consistent Tagging in One Click

Bundle Recurring Tag Sets

TAG COLLECTION **OSINT Report**

- tlp:clear
- workflow:state="complete"
- osint:source-type="blog-post"
- admiralty-scale:source-reliability="c"
- estimative-language:confidence-in-analytic-judgment="moderate"
- misp-galaxy:sector="Finance"

Bundle
Combine taxonomy tags and galaxy clusters into a named collection.

Apply
Assign a collection at event or attribute level in one click.

Share
Use privately or share with communities and exchange as JSON.

Standardise
Consistent classification across the team, fewer forgotten tags.

Tagging in Practice

Local or Global, Event or Attribute

	Global Tag	Local Tag
Synchronised	Yes, to all recipients	No, stays on your instance
Typical Use	Shared classification (TLP, galaxies)	Internal routing, own assessment
Example	tlp:amber	workflow:todo="review"

EVENT VS. ATTRIBUTE CONTEXT

- Event tags are not copied onto each attribute
- Exports can include event context, which differs from an attribute tag
- Tag attributes only where the value needs its own assessment

Three Tag Settings

local_only
The tag can only be attached as a local tag.

exportable
Whether the tag may appear in exports; other rules still apply.

Org / User Restriction
Who may apply the tag, not who may read the event.

Remember: Restricting tags never replaces distribution.

Rating Sources with Tags

Source Reliability and Information Credibility

	Source Reliability		Information Credibility
A	Completely reliable	1	Confirmed by other sources
B	Usually reliable	2	Probably true
C	Fairly reliable	3	Possibly true
D	Not usually reliable	4	Doubtful
E	Unreliable	5	Improbable
F	Cannot be judged	6	Cannot be judged

Admiralty Scale: e.g. "B2" = usually reliable source, probably true.

As Tags in MISP

```
admiralty-scale:source-reliability="b"
admiralty-scale:information-credibility="2"
```

Travels with the Data

The rating is synced and can be filtered in searches and exports.

More Nuance

estimative-language adds likelihood and analytic confidence.

Tip: Rate the source once per feed or partner and the content per event. Decaying models can use these tags for the base score.

Let's Take a Look in MISP

Live Demo: Tagging in Practice



WHAT WE WILL LOOK AT

1. Taxonomies: which ones are enabled on our instance
2. Tag collections: apply a recurring tag set in one click
3. Global or local tag: what is synced and what stays here
4. Rate source and content with admiralty-scale tags
5. Filter events by tag in the index and via restSearch

LIVE DEMO: APPROX. 10 MINUTES



Feeds: Connect External Sources

MISP as a Data Sink for OSINT and Commercial Sources

Caching
Cache a feed locally and correlate on hits only, without creating events.

Fetch & Store
Import feed content as events, incl. default tags.

Overlap Analysis
Compare feeds and identify redundancies.

Everyday Context
Hits on Tor exit nodes, sinkholes or known IoCs become visible immediately.

Format	Description	Example
MISP Feed	Events in MISP JSON format incl. context	CIRCL OSINT Feed
CSV	Column-based indicator lists	abuse.ch lists
Freetext	Unstructured lists, MISP extracts values	IP or URL blocklists
Custom Feed	Generated or converted with PyMISP	Internal IoC collection

Tip: Only cache new feeds at first and watch the correlations before importing them as events.

Integrating OSINT Feeds into Detection

From Free Feed to Reviewed Detection Content

Select	Ingest	Qualify	Deliver	Review
Relevance, licence and quality of the source	Cache or fetch; default tags such as osint, tlp:clear	Warninglists, short decay lifetime, correlation	restSearch with tags, to_ids, excludeDecayed	Sightings and FP rate per feed; drop weak feeds

Feed (Examples)	Content	Use in Detection
CIRCL OSINT Feed	Events from public reports	Hunting, correlation
abuse.ch URLhaus / ThreatFox	Malware URLs and IoCs	Proxy and DNS blocking after review
abuse.ch Feodo Tracker	Botnet C2 IP addresses	Firewall blocking, short lifetime
Tor exit nodes	Anonymisation infrastructure	Context only, no blocking

Keep in mind: OSINT supplements, it does not replace curated sources. Tag each feed on ingest so you can filter it and measure its quality.

Default feeds: Sync Actions › Feeds › Load default feed metadata. Check licence and terms of use for each feed.

MITRE ATT&CK in MISP

Describe Attacker Behaviour Consistently

Tag Techniques

Link events and attributes to ATT&CK techniques, e.g. T1566 Phishing.

Matrix View

Show the techniques of an event or a selection as a heatmap.

Export

Export as an ATT&CK Navigator layer and compare with your detection coverage.

Bridge to Detection

TTPs help prioritise use cases and hunting hypotheses.

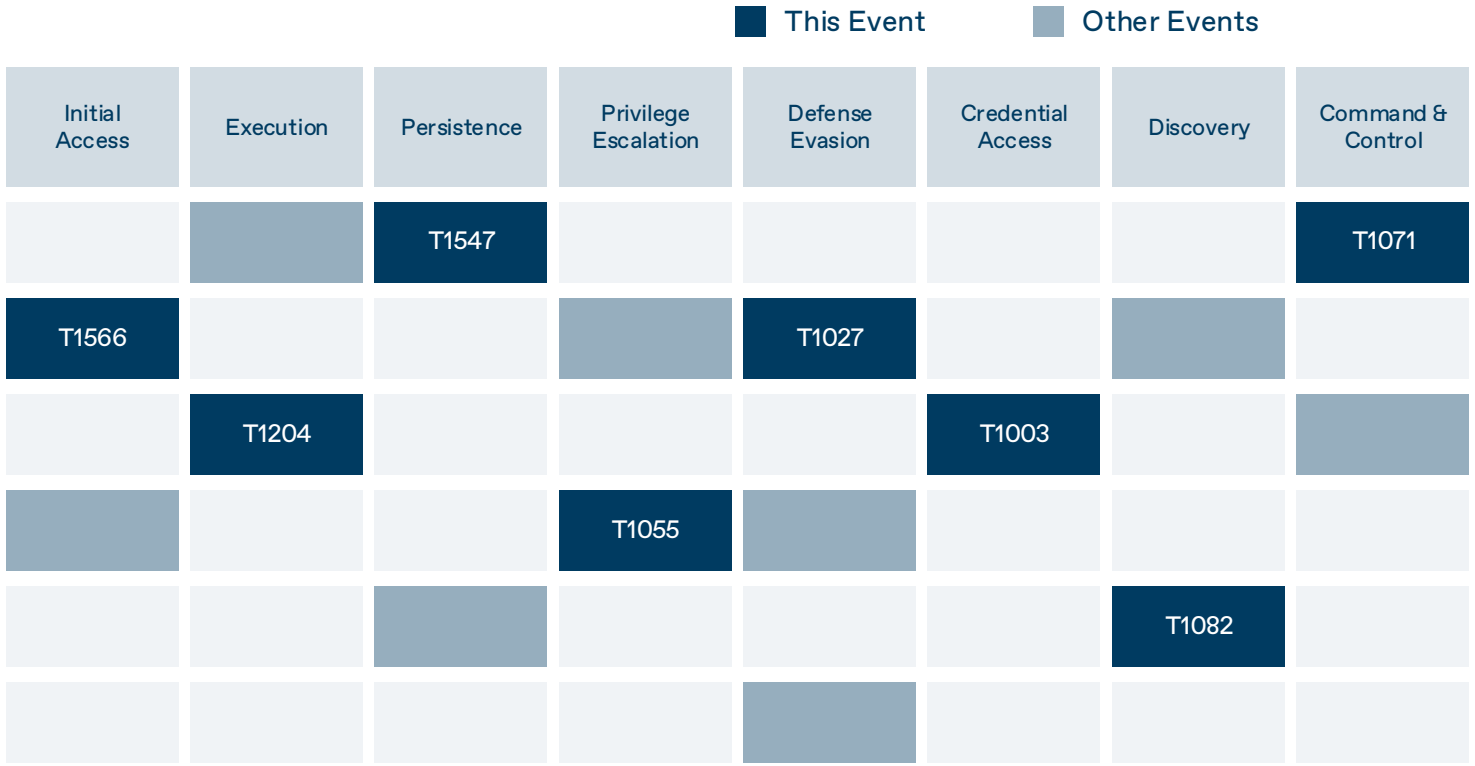


Illustration: ATT&CK matrix view with the techniques of one event.

Query example via the API on the next slide.

ATT&CK Heatmaps via the API

Which Techniques Target Our Sector?

REST API

```
POST /events/restSearch
{
  "returnFormat": "attack",
  "tags": ["misp-galaxy:sector=\"Finance\""],
  "timestamp": "365d"
}
```

PYMISP

```
matrix = misp.search(controller="events",
    return_format="attack",
    tags=['misp-galaxy:sector="Finance"'],
    timestamp="365d")
```

What It Returns

An ATT&CK matrix with how often each technique appears in the matching events.

Typical Questions

Which techniques hit our sector this year? Which actors use them?

Filters

Combine tags, organisations, published status and time frame as usual.

Next Step

Compare the heatmap with your detection coverage and close gaps.

Let's Take a Look in MISP

Live Demo: Feeds and the ATT&CK Heatmap



WHAT WE WILL LOOK AT

1. Feed overview: default feeds and how to enable one
2. Fetch a feed: cache it or import it as events
3. Feed hits: how a cached feed correlates with our events
4. ATT&CK matrix of a single event
5. Heatmap via the API: restSearch with returnFormat attack

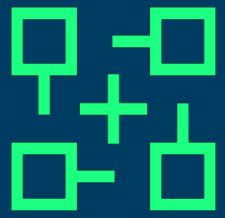
LIVE DEMO: APPROX. 10 MINUTES



Back to the Data Model

A Deeper Look at Events

In chapter 1 we met events, objects and attributes. Now that we have worked with them, we look at what this design means in daily work.



What We Already Know

Events Are Containers

Everything about one incident or report lives in one event.

Attributes and Objects

Single values and structured groups of values with references.

Context Around It

Tags, galaxies and correlations connect events with each other.

Now the Deep Dive

Event-Based vs. IOC-Based

Two ways to model threat data

Operational Cost

Where the event model reaches its limits

Event-Based vs. IOC-Based

What MISP's Data Model Means for Daily Work

Event-Based How MISP works

Indicators live in the context of an incident, report or analysis.

+ Context is preserved: who, when, why, which campaign
+ Traceable back to the original source
+ Natural unit for sharing, publishing and assessing
+ Correlations reveal links between events
– The same value appears in many events (duplicates)
– No central assessment per indicator
– Lifecycle of single IoCs requires decaying and sightings
– High operational cost with large data volumes (next slide)

IOC-Based e.g. indicator feeds and platforms

Each indicator is a standalone object with a score and expiry.

+ One record per value with a single assessment
+ Fast, simple distribution to SIEM, EDR and firewall
+ Expiry and score directly controllable per indicator
+ Optimised for bulk queries and large lists
– Context is easily lost: why is the value malicious?
– Links to campaigns and actors hard to see
– Analysts must research the background separately

Best of both worlds: MISP provides event context and serves the IOC view via attribute search, decaying, sightings and flat export lists.

Operational Cost of the Event Model

Where MISP Reaches Its Limits with Large Data Volumes

Searching Large Data Volumes

- Queries across millions of attributes take long and tie up RAM and CPU
- Every context parameter (tags, galaxies, includeContext, decay score) adds load
- Substring searches and frequent SIEM queries quickly lead to timeouts

Large Lists of Plain Indicators

- The anchor is the event, not the IoC: every value becomes an attribute of an event
- Huge collection events are slow to view, sync and export
- Correlations and tags make the database grow fast

Keeping Costs Under Control

- Flat export lists or caches for the SIEM (two-step matching)
- API proxy or dedicated indicator API for bulk queries
- Only cache feeds instead of importing them as events
- Disable correlation for bulk data
- Tags on the event rather than on every attribute
- Limit and rotate collection events

Key Takeaways · Chapter 3

Deep Dive: Working with MISP



WHAT WE HAVE LEARNED

- [ ] Keep noise out with warninglists and blocklists, and know which built-in lists to enable
- [ ] Decide whether to cache a feed or import it as events
- [ ] Add context with MITRE ATT&CK and tag collections
- [ ] Tell local from global tags and source reliability from information credibility
- [ ] Weigh event-based against IOC-based data, including the operational cost

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



4 MISP Use Cases

From Use Case to Detection

MISP in Action: Five Proven Use Cases

Same Building Blocks, Different Topology

I know how MISP works, but now what?

The following examples show how versatile MISP is.



1 Content Aggregation & Enrichment

MISP bundles sources; SIEM hits are enriched with context in two steps.



2 Active Measures via Tags

Tags decide which indicators are automatically blocked at the firewall or proxy.



3 Feed Funnel & Qualification

MISP as a quality gate filters feeds before they reach production systems.



4 Incident Response Cooperation

Context from MISP supports case management, e.g. in TheHive.

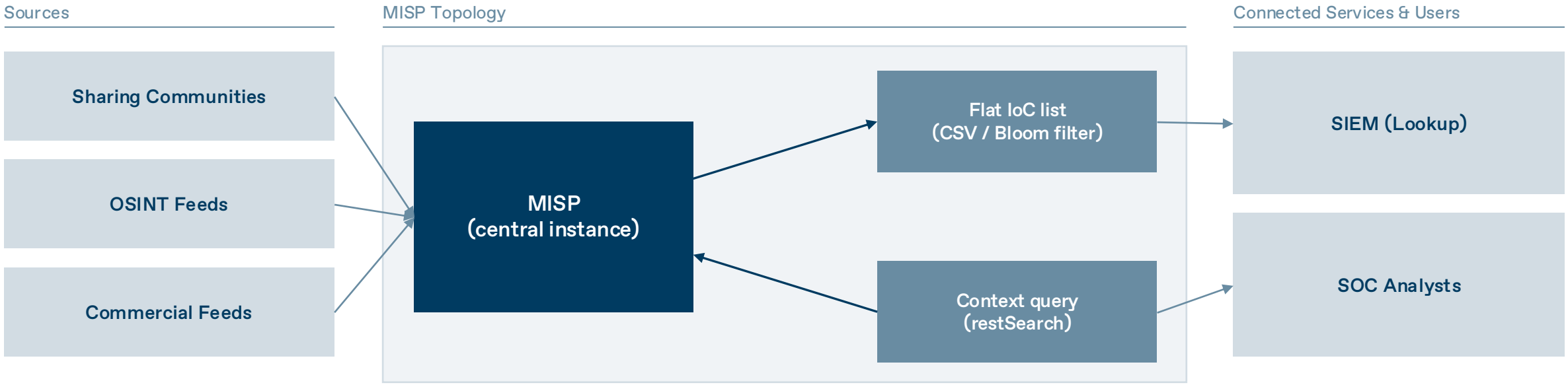


5 Feed Provider Model

Multiple instances supply recipient groups with appropriately classified data.

Use Case 1: Content Aggregation & Enrichment

Fast Matching in the SIEM, Full Context for Analysts



Key Idea

MISP bundles all sources. The SIEM matches against a flat list; context is fetched for hits only.

Benefits

Low load on MISP, fast lookups in the SIEM, full event details when an alert fires.

Watch Out For

Define the list refresh interval; avoid direct SIEM queries against MISP.

Two-Step Matching for Large SIEMs

Relieve MISP Without Losing Context

Step 1: Lookup

Flat IoC List from MISP

Export regularly as CSV or Bloom filter and use it as a lookup table in the SIEM.

```
POST /attributes/restSearch
{"returnFormat": "csv",
 "type": "ip-dst", "to_ids": 1,
 "publish_timestamp": "30d"}
```

Step 2: Enrichment

Context for Hits Only

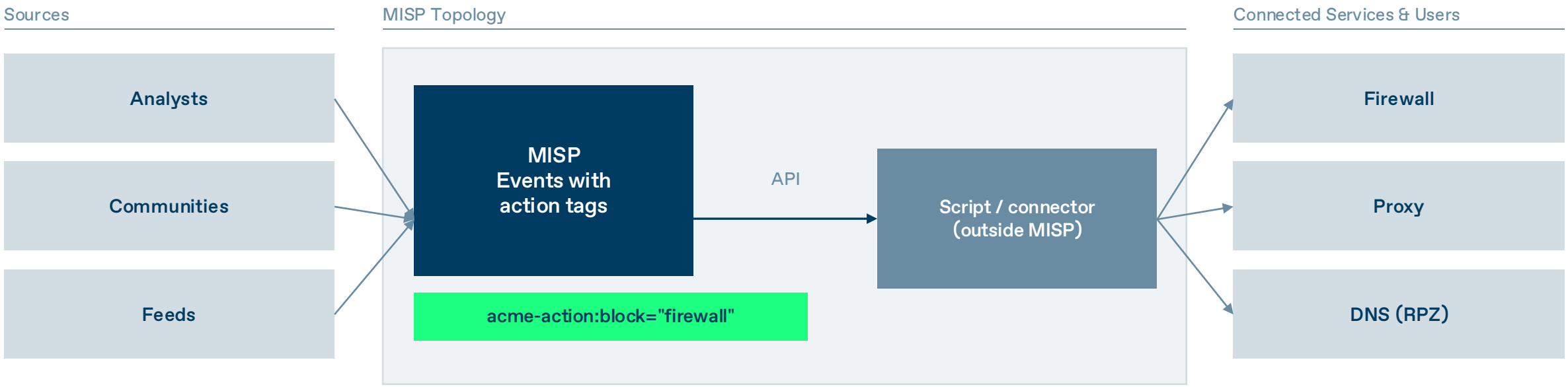
On a hit, query the value in MISP and give analysts the event, tags and galaxies.

```
POST /attributes/restSearch
{"returnFormat": "json",
 "value": "203.0.113.10",
 "includeContext": 1}
```

Result: The SIEM matches quickly against a local list; MISP answers only a few targeted queries.

Use Case 2: Active Measures via Tags

Indicators to Perimeter Systems Without Copy & Paste



Key Idea

Tags such as acme-action:block="firewall" define which indicators trigger an action.

Benefits

No manual transfer, traceable via tags and event history.

Watch Out For

Tag at event level, enforce warninglists, define an approval process.

Active Measures via Tags

Indicators to Perimeter Systems Without Copy & Paste

1 Tag

An analyst marks an event, e.g. with acme-action:block="firewall".

2 Query

A script fetches all attributes with this tag via restSearch.

3 Enforce

The list is written automatically to firewall, proxy or DNS RPZ.

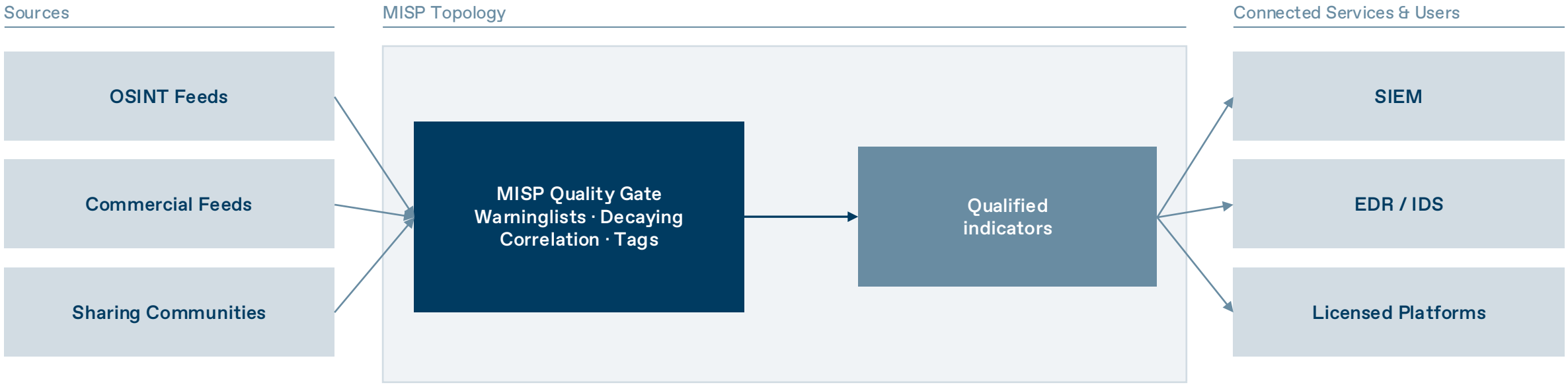
```
POST /attributes/restSearch
{"returnFormat": "text", "type": "ip-dst",
 "tags": ["acme-action:block=\"firewall\""],
 "to_ids": 1, "enforceWarninglist": 1}
```

Important: Set tags at event level. Many attribute tags significantly increase loading times.

acme-action is an example of a custom taxonomy following the schema [identifier]-[namespace]:predicate="value".

Use Case 3: Feed Funnel & Qualification

Only Reviewed Indicators Reach Production Systems



Key Idea

MISP filters incoming feeds before they reach production or volume-licensed systems.

Benefits

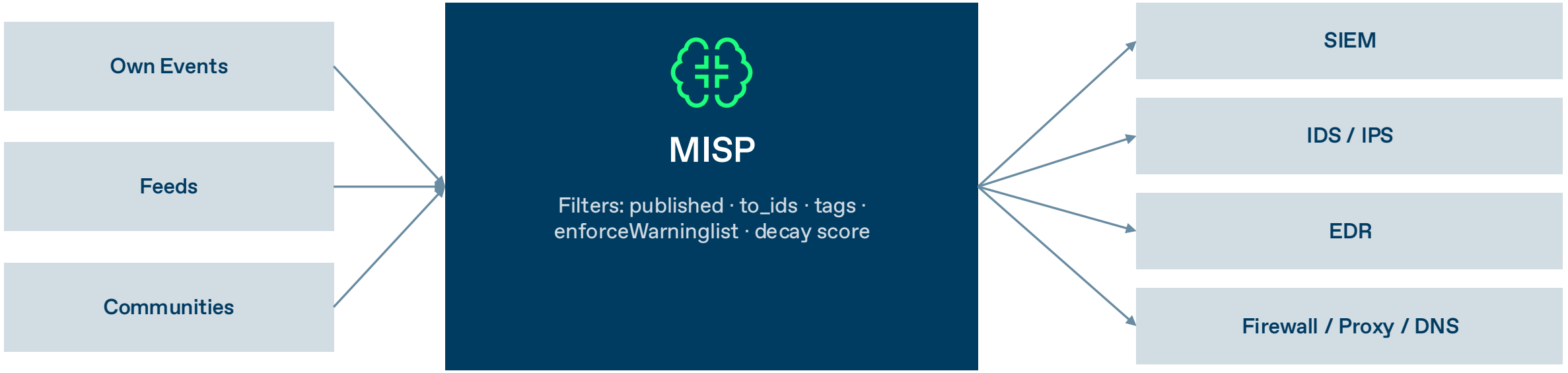
Fewer false positives, lower licence costs, import with almost no loss of information.

Watch Out For

Document filter rules and review them regularly against hit and FP rates.

From Event to Detection

Feed Indicators into Security Systems Automatically



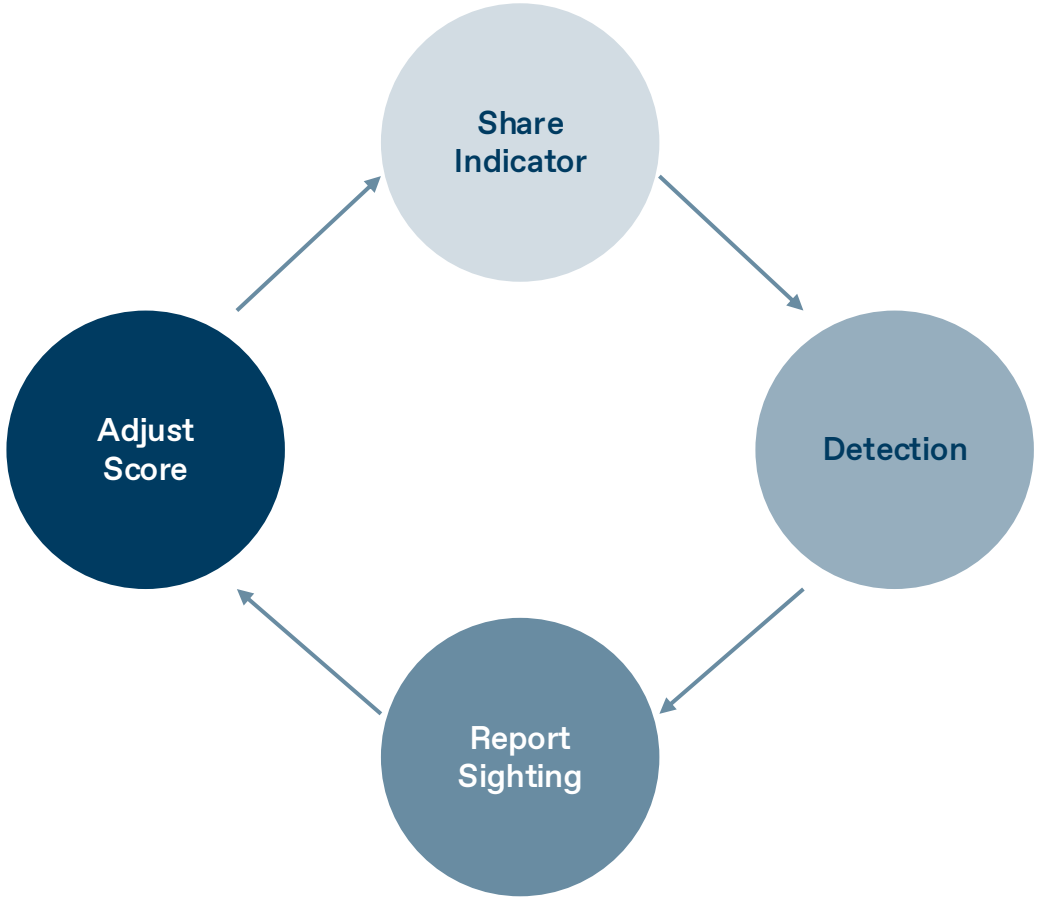
Pull model: Systems fetch indicators regularly via restSearch.

Push model: ZeroMQ, Kafka or workflows report changes in real time.

Feedback: Report hits back as sightings and assess indicators.

Safeguarding Detection Quality

Decaying Models and Sightings as a Feedback Loop



Decaying Models

Compute a score that drops as an indicator ages, depending on type and tags.

Sightings

Feedback: value seen, false positive or expired.

Export by Score

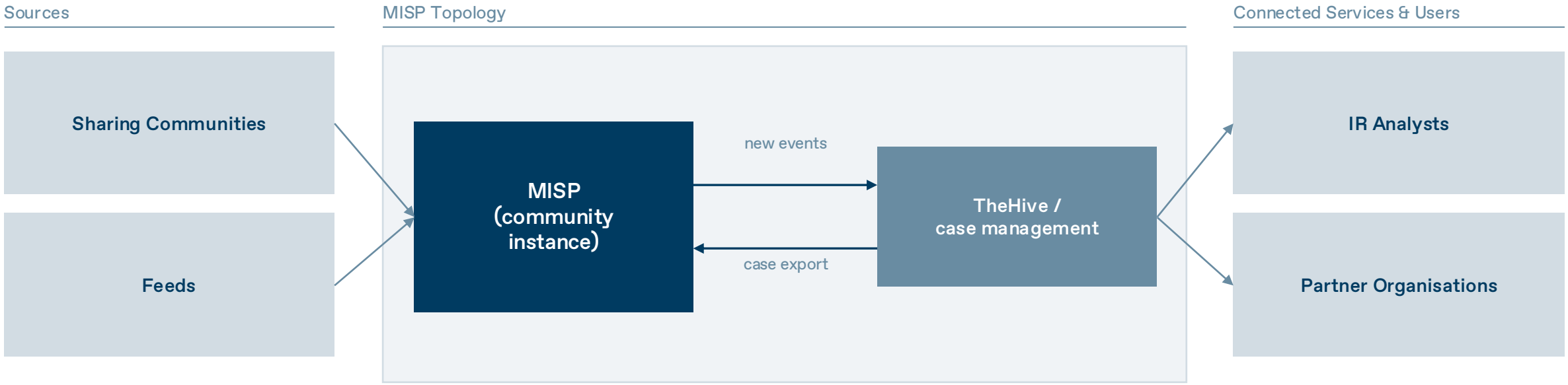
restSearch with excludeDecayed returns only indicators above the threshold.

Limit Lifetime

First/last seen and expiration sightings end the use of outdated values.

Use Case 4: Incident Response Cooperation

Context from MISP for Faster Decisions in Case Management



Key Idea

Case management queries MISP for new events; cases flow back as events.

Benefits

Fast triage of observables, built-in TheHive integration, low load.

Watch Out For

Choose the distribution deliberately when exporting from a case.

Example: TheHive and Velociraptor

From Shared Event to Hunt and Back

1 Alert	2 Case	3 Hunt	4 Findings	5 Share Back
TheHive imports a new MISP event as an alert via its MISP connector.	The analyst opens a case and enriches observables, e.g. with Cortex.	Velociraptor pulls hashes, YARA rules and domains from MISP and hunts on all endpoints.	Hits and new indicators from the hunt go into the case.	Sightings for confirmed hits, the case exported to MISP as a new event.

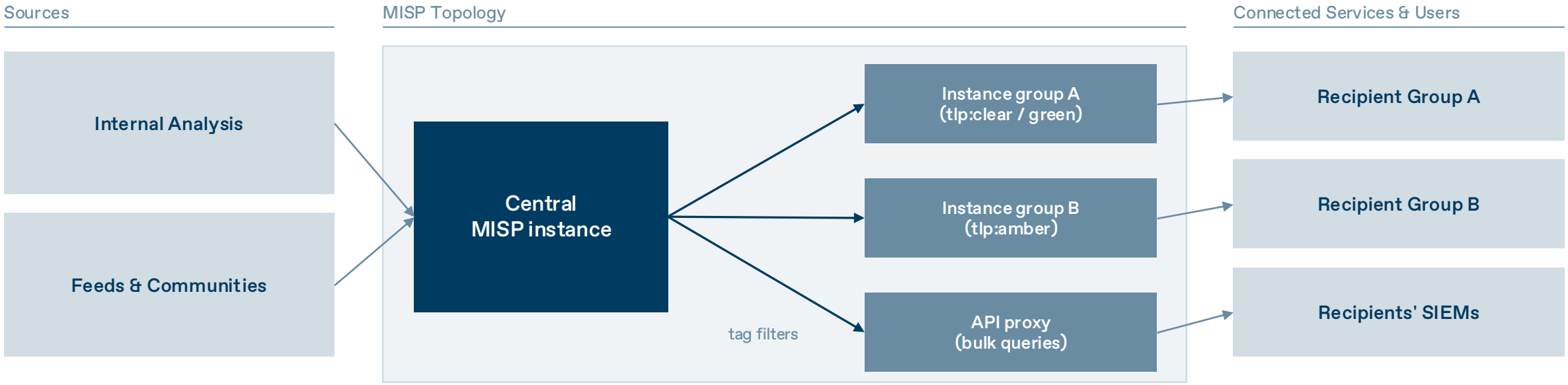
TheHive: Built-in MISP connector in both directions: events in as alerts, cases out as events.

Velociraptor: No native connector. A small script or server artifact queries restSearch with to_ids and tags.

Works the same with other case management and DFIR tools: MISP stays the shared source, the tools only consume and report back.

Use Case 5: Feed Provider Model

Every Recipient Group Gets Exactly the Right Data



Key Idea

Multiple instances separate recipient groups; tags and filters steer the data flow.

Benefits

Clear separation by TLP and contract, scalable supply to many organisations.

Watch Out For

Plan an API proxy for high-performance SIEMs, as the MISP API is rather slow.

Export Formats & Integrations

The Right Format for Every System

Target System	Format / Method	returnFormat
SIEM (Splunk, Elastic, Sentinel, QRadar)	JSON or CSV via restSearch, apps & connectors	json, csv
IDS / IPS (Suricata, Snort)	Rule export	suricata, snort
DNS firewall	Response Policy Zone	rpz
Firewall / proxy	Text lists (IP, domain, URL)	text
EDR / malware scanning	Hash lists, YARA rules	hashes, yara
TIPs and partners	STIX 2.x	stix2
ATT&CK Navigator	Layer with techniques	attack

Common filters: published, to_ids, tags, last / publish_timestamp, enforceWarninglist, includeDecayScore, excludeDecayed.

Key Takeaways · Chapter 4

MISP Use Cases



WHAT WE HAVE LEARNED

- [ ] Name the five use cases and pick a topology that fits your sources, services and users
- [ ] Filter indicators for detection: published, IDS flag, warninglists, decay score
- [ ] Use two-step matching instead of direct SIEM queries against MISP
- [ ] Trigger active measures on perimeter systems with tags
- [ ] Bring OSINT feeds into detection and close the loop with sightings
- [ ] Choose the right export format for each target system

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



5 Connecting MISP

MISP API & PyMISP

MISP REST API

Almost Everything the UI Can Do, Automated

JSON over HTTPS, authenticated with the auth key in the Authorization header.

```
curl -X POST https://misp.example/attributes/restSearch \
  -H "Authorization: <AUTH_KEY>" \
  -H "Accept: application/json" \
  -H "Content-Type: application/json" \
  -d '{"returnFormat": "csv", "type": "ip-dst",
      "to_ids": 1, "last": "7d"}'
```

Security: Dedicated API user with a minimal role; auth keys with expiry date and IP allowlist.

Endpoint	Purpose
/events/restSearch	Search and export events
/attributes/restSearch	Filter attributes, detection exports
/events/add	Create an event
/attributes/add/{event_id}	Add an attribute
/events/publish/{event_id}	Publish an event
/sightings/add	Report a sighting
/tags/attachTagToObject	Attach a tag to an event or attribute

All endpoints and parameters: the instance's built-in REST client and OpenAPI docs.

PyMISP: MISP in Python

The Official Library for Automation

```
from pymisp import PyMISP, MISPEvent

misp = PyMISP("https://misp.example", AUTH_KEY, ssl=True)

# Fetch indicators from the last 7 days
attrs = misp.search(controller="attributes",
                    type_attribute="ip-dst", to_ids=True,
                    last="7d", pythonify=True)

# Create a new event
event = MISPEvent()
event.info = "Phishing wave October 2026"
event.distribution = 0
event.add_tag("tlp:amber")
event.add_attribute("domain", "invoice-check.example", to_ids=True)
misp.add_event(event)
```

Object Model

MISPEvent, MISPObject and MISPAttribute map the data model to Python.

Search & Export

search() wraps restSearch with all filters and formats.

Generate Feeds

Generate your own MISP feeds or convert third-party data.

Installation

pip install pymisp; keep the version in line with your MISP instance.

Key Takeaways · Chapter 5

Connecting MISP



WHAT WE HAVE LEARNED

- [ ] Query MISP via the REST API and restSearch filters
- [ ] Automate recurring tasks with PyMISP
- [ ] Pick the right endpoint to search, create and publish events
- [ ] Secure API access with dedicated users and expiring auth keys

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.

6 Operating MISP

Options for Running Your Own Instance

Operating Models Compared

Finding the Right Path to Your Own Instance

Criterion	Native Installation	Docker (misp-docker)	Managed MISP
Setup effort	Medium, install scripts for Ubuntu and RHEL	Low, docker compose	Very low
Control	Full	High	Limited by provider
Updates	Manual via Git or UI	Pull a new image	By provider
Know-how	Linux, web server, database	Container operations	Low
Features	All, incl. ZeroMQ & workflows	All	Depends on offering
Suited for	Full customisation, air-gap	Quick start, test & production	Focus on use, not operation

Further options according to CIRCL: MISP VM images, RPM packages, cloud images (misp-cloud) and LXC images for air-gapped environments (misp-airgap).

Sizing and Performance

What Really Drives an Instance's Performance

What Drives the Load

- Correlations (RAM, disk speed, storage)
- Feed ingestion strategy
- Over-contextualisation: tags and galaxies better on the event than on every attribute
- Concurrent active users and heavy API searches (e.g. substring)
- Logging strategy (disk space)

Less Critical than Expected

- Warninglists
- Raw number of attributes
- Number of sync connections (within reason)
- Tools on ZeroMQ, Kafka or syslog

Hardware: SSDs make the biggest difference. No firm requirements; plan a bit above what you need.

Authentication: Password by default; modules for LDAP, OpenID Connect, Azure AD, Shibboleth, x509 and e-mail OTP.

High Availability and Tuning

When One Server Is No Longer Enough

High Availability

- Load-balanced web servers running MISP
- Replicated or mirrored database back end
- Watch session pinning on the load balancer
- Shared or network-attached attachment storage
- AWS example: github.com/0xtf/HAMISPA

Tuning

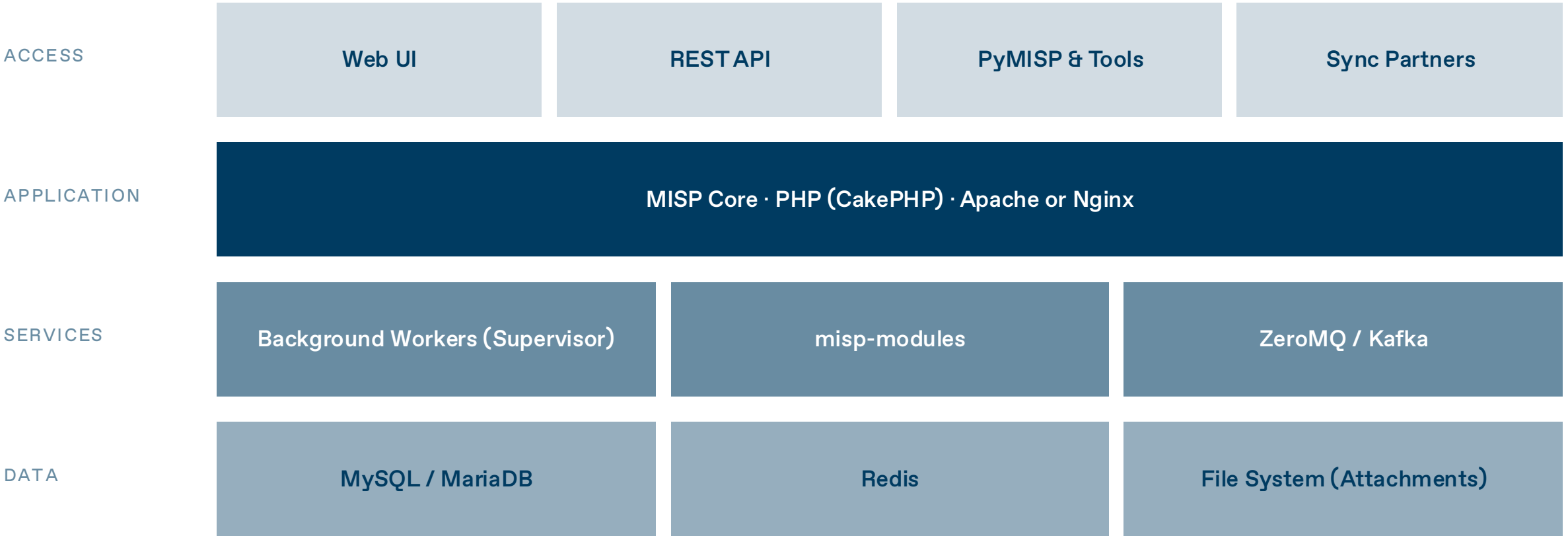
- PHP memory limit and timeouts, configurable per role
- Background workers are exempt from web limits
- MySQL: key buffer size is important
- Tune for few heavy requests, not many light ones
- Redis for caching and job queues

Before clustering: Check correlations, feed strategy and attribute tags first. These often cost more than missing hardware.



Architecture of a MISP Instance

Components that Matter in Operation



Logs: application logs in the database, server logs under /var/www/MISP/app/tmp/logs/.

Day-to-Day Administration

Lessons from Operations



Workers & Jobs

Workers fail occasionally, especially after updates. Plan a health-check script with auto-restart.



Scheduled Tasks

Trigger pull and push via cron and the CakePHP console if the built-in tasks hang.



Diagnostics

Checks dependencies and settings, including a security audit.



Logs

Database logs grow steadily and slow the instance down; clean them up manually on a regular basis.



Mail Encryption

GnuPG or S/MIME per user; setting up S/MIME is laborious.



Roles & Permissions

Use the role concept: Org Admin, Publisher, User, Sync User, Read Only.



Users, Organisations and Roles

Set Up Access Cleanly from the Start

Role	Can Do	Typical For
Site Admin	Everything on the instance	Platform operators
Org Admin	Manage users of own organisation, publish	Team leads
Publisher	Create, edit and publish events	Senior analysts
User	Create and edit events, no publishing	Analysts
Read Only	View and search, no changes	Consumers, management
Sync User	Synchronise with partner instances	Server connections

Organisations First

Create local and remote orgs with the correct UUID.

Custom Roles

Define roles with exactly the permissions needed.

API Users

Separate users with auth key expiry and IP allowlist.

Tip: Use templates for welcome e-mails and set default distribution per organisation.

Operations Checklist

1

Updates & Versions

Update regularly, incl. taxonomies and warninglists.

2

Backup & Restore

Back up database, configuration and attachments.

3

Hardening

TLS, MFA or SSO, minimal roles and auth keys with expiry.

4

Monitoring

Monitor workers, disk space and error logs.

5

Data Quality

Maintain feeds, warninglists and sync filters.

6

Governance

Document roles, sharing rules and responsibilities.

TLP:CLEAR

Key Takeaways · Chapter 6

Operating MISP



WHAT WE HAVE LEARNED

- [ ] Compare native, Docker and managed operation
- [ ] Name what really drives load and sizing
- [ ] Know the components of an instance and what to monitor
- [ ] Set up users, organisations and roles cleanly
- [ ] Use the operations checklist for updates, backup and hardening

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



7 Stretch Topics

Extra Material if Time Allows

Why Decaying Matters

Old Indicators Cause False Positives, Load and Manual Work

Without Decaying

- Export lists only ever grow
- Reassigned IPs and domains trigger false positives
- Every indicator is treated the same, old or new
- Manual clean-up of outdated values

With Decaying

- Stale indicators drop out of exports automatically
- Lifetime fits the type: short for IPs, long for hashes
- Score reflects age, source confidence and sightings
- Consumers act by score: block, alert or ignore

Example: IP Address from a Phishing Campaign (Illustrative Values)

Day 1 · Score 80	Day 20 · Score 45	Day 35 · Score 15
Fresh and confirmed: blocked at the proxy	Older, no new sightings: alert only, no block	Below threshold: removed from the export

A new sighting at any point resets the clock and lifts the score again.

Decaying Models in Detail

How MISP Scores the Age of an Indicator

$$score(t) = base_score \times (1 - (t / \tau)^{(1 / \delta)})$$

base_score	From taxonomy tags with weights and the source organisation's confidence.
τ Lifetime	Time until the score reaches zero, e.g. 30 days for IPs, longer for hashes.
δ Decay Speed	Shape of the curve: fast drop at the start or late drop before expiry.
Threshold	Below this score the indicator counts as decayed and can be excluded.
Sightings	A new sighting resets the clock, so active indicators stay relevant.

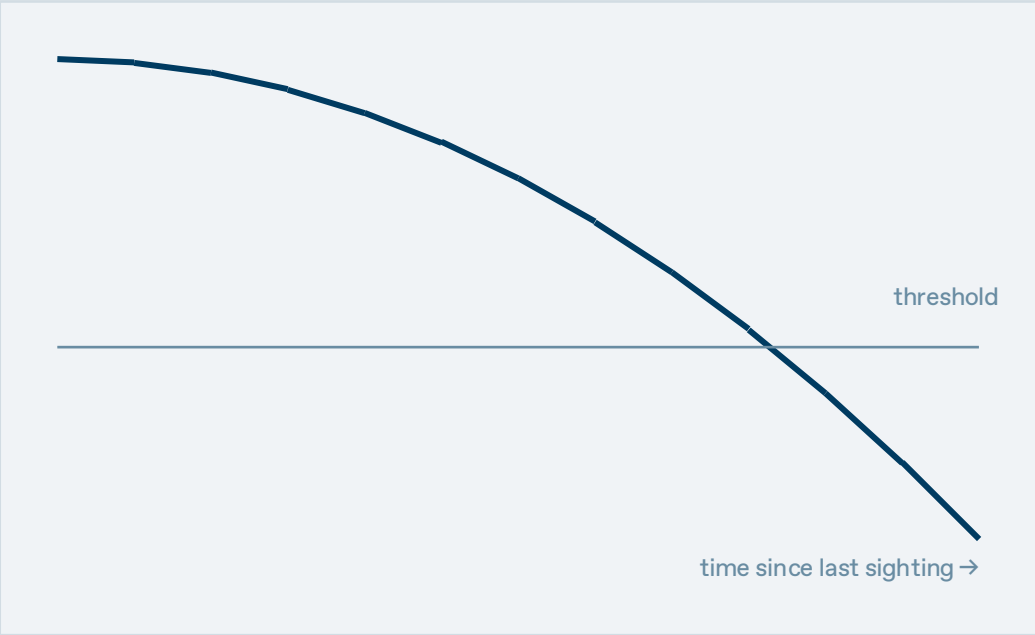


Illustration: score drops over the lifetime τ

API: restSearch with includeDecayScore and excludeDecayed.

Event Reports: Narrative and Data Together

Write the Analysis Where the Indicators Live

EVENT REPORT **Phishing Wave October**

Summary

The campaign uses fake invoice portals. Victims are redirected to @[attribute](invoice-check.example).

Infrastructure

C2 traffic goes to @[object](domain-ip). The actor overlaps with @[galaxymatrix](T1566).

Linked elements stay clickable and are updated automatically.

Markdown in the Event

Write analysis, timelines and recommendations next to the data.

Linked References

Mention attributes, objects and galaxy clusters directly in the text.

Entity Extraction

MISP suggests indicators from the text and converts them into attributes.

Own Distribution

Each report can be shared more narrowly or widely than the event.

Event Templates: Consistent Events from the Start

New 2026: Rewritten Templating System

EVENT TEMPLATE **Spearphishing**

Preset Tags	tlp:amber · workflow:state="incomplete"
Galaxy	mitre-attack-pattern: Spearphishing Link - T1566.002
Objects	email · url · file
Required Fields	email subject, sender, at least one URL or file

STARTER TEMPLATES (EXAMPLES)

Ransomware | **Credential Exposure** | **Spearphishing** | **Supply Chain**

Builder UI

Drag and drop with inline validation and a live preview.

Pickers

Choose object templates, taxonomies and galaxies directly in the builder.

Templates Library

Ten ready-to-use starter templates for common cases.

Migration Helper

Converts legacy templates to the new system.

Benefit: Every analyst captures the same fields and context, so events are complete and comparable.

Dashboards: Monitor Trends in MISP

Build Views that Support Decisions

Analyst View

- Trending tags and galaxies
- New events from key communities
- Sightings and false positives
- ATT&CK coverage of recent events

Operator View

- Worker and job status
- Sync and feed health
- Users and organisations
- Instance usage over time

Widget Gallery
Choose widgets by question, configure with preview.

Templates
Start from the template library and share layouts.

Shared Filters
One filter drives several widgets; keep the scope visible.

Maps
New 2026: geolocation and map visualisation.

Watch out: Refresh rate is not data freshness, and users may see different numbers because of distribution and visibility.

misp-modules: Enrich, Import, Export

Extensions Without Changing the MISP Core

Enrichment

Enrich attributes in one click, e.g. with passive DNS, WHOIS or VirusTotal.

Import

Bring data from formats such as e-mail, OpenIOC or CSV into events.

Export

Convert events into further formats or reports.

Separate Python Service
misp-modules runs as a separate server; MISP talks to it via HTTP.

Easy to Enable
Configure modules under Server Settings › Plugin Settings; MISP detects new modules automatically.

Watch out: No native TLS and no proxy support, so bind the service locally or secure it via a tunnel. Python 3 and Git are required.

MISP Workflows

Automation Directly in MISP



Predefined by MISP, e.g. tag added or event published.

Conditions that must be met. Chainable, with filters, also in parallel.

Runs when the condition is met: from notification to blocking.

Caution: Workflows can block internal actions such as publishing. Only one workflow per trigger is possible; test thoroughly before production use.

Editor directly in MISP under Administration › Workflows; misp-modules must be running.

Workflow Use Cases

What Teams Automate with MISP Workflows

Auto Submission

Trigger: New malware sample or file attribute

Action: Submit to the sandbox and attach the report

Chat Notification

Trigger: Event published with a relevant tag

Action: Post a message to Slack, Mattermost or Teams

Set Workflow Tags

Trigger: New event arrives via feed or sync

Action: Add workflow:state="incomplete" for review

Stop Sensitive Publishing

Trigger: Event publish with tlp:red content

Action: Block publishing and inform the author

Auto Enrichment

Trigger: Domain or IP attribute added

Action: Run passive DNS or WHOIS and add results

Quality Checks

Trigger: Attribute hits a warninglist

Action: Remove the IDS flag and tag it for review

Workflows in Detail

Triggers, Blocking Behaviour and Blueprints

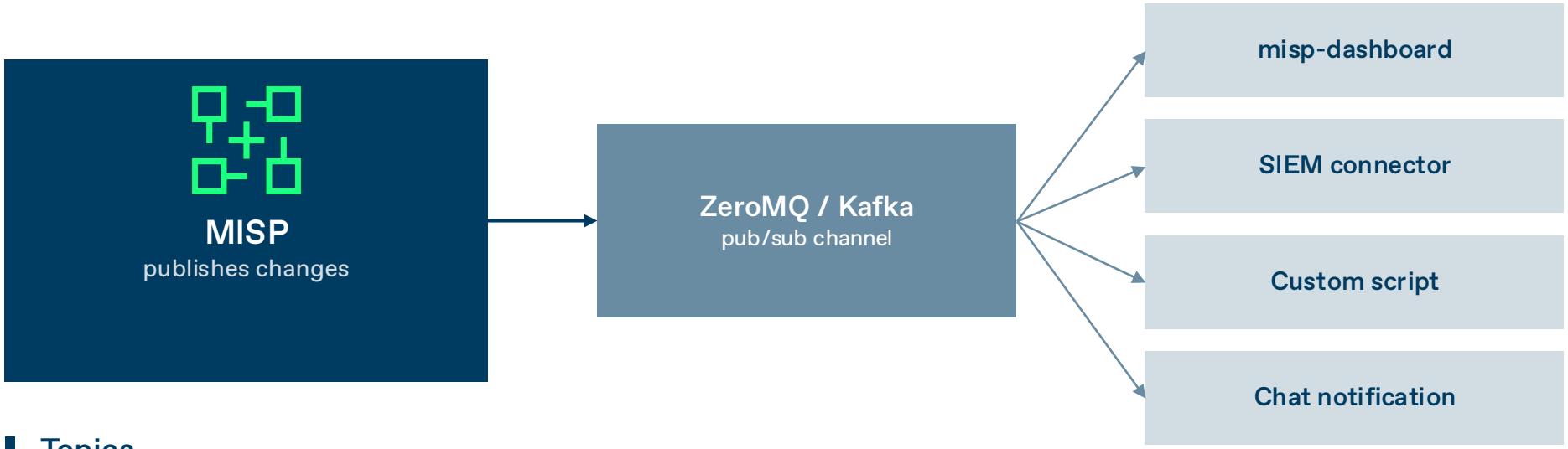
Aspect	Details
Triggers	Event publish, attribute/object changes, sighting added, user login and more
Blocking	Can stop the triggering action, e.g. block publishing if tlp:red is present
Non-blocking	Runs after the action, e.g. send a notification or enrich
Logic Modules	If/else, filters, concurrent tasks, stop execution
Action Modules	Tag, enrich, notify (mail, Mattermost, webhook), push to ZeroMQ
Module Sources	MISP core, misp-modules and custom Python modules
Blueprints	Reusable workflow fragments that can be shared

Example: Blocking workflow on event publish that stops publishing when an attribute carries tlp:red.

Tip: Start with non-blocking workflows and test blocking ones on a staging instance first.

Real-Time Integration: ZeroMQ and Kafka

React to Changes Instead of Polling



Topics
misp_json_event, misp_json_attribute, misp_json_sighting, misp_json_tag and more.

Push Instead of Pull
Consumers react within seconds; no repeated restSearch queries against MISP.

Mind the Limits
Not every managed offering exposes ZeroMQ. Schema: circl.lu/doc/misp/misp-zmq.

AI and MISP: LLMs, Agents and Claude

Current Building Blocks and How to Use Them Safely

misp-mcp

MCP server from the MISP project; read-only access for assistants such as Claude

ai-connector

misp-module for local LLMs via Ollama, vLLM or LM Studio

information-origin

Taxonomy to mark AI-generated content; ai-chat-prompt object to share prompts

AI-Assisted Development

CIRCL guide to building misp-modules with an AI agent

Typical Use Cases

- Ask MISP in natural language: "Which events mention APT28 this month?"
- Summarise events or draft event reports for management
- Extract indicators and ATT&CK techniques from report text
- Prepare enrichment and correlation analysis for analysts

Guardrails

- TLP still applies: no AMBER or RED data to public AI services
- Prefer local or contractually approved models
- Read-only access first; a human reviews before publishing
- Mark AI-generated content with information-origin

Key Takeaways · Chapter 7

Stretch Topics



WHAT WE HAVE LEARNED

- [ ] Write event reports that link narrative and data
- [ ] Use event templates for consistent, complete events
- [ ] Explain how decaying models score the age of an indicator
- [ ] Build dashboards for analysts and operators
- [ ] Extend MISP with misp-modules for enrichment, import and export
- [ ] Judge when workflows help and where they can block actions
- [ ] Use ZeroMQ or Kafka for real-time integration
- [ ] Use AI assistants with MISP safely, e.g. via misp-mcp

QUICK CHECK: ANYTHING UNCLEAR? NOW IS THE TIME TO ASK.



Key Takeaways of the Day

What to Remember from Today



Structure

Data model, taxonomies and galaxies make information machine-readable.



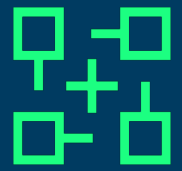
Share

Distribution, sharing groups and filters control who sees what.



Ensure Quality

Warninglists, IDS flag, decaying and sightings keep indicators usable.



Automate

REST API, PyMISP and workflows connect MISP to detection.

Further Resources

For Reference and Deeper Learning

Resource	Content	Link
MISP Book	Reference documentation	circl.lu/doc/misp
MISP Cheat Sheet	Compact overview for daily work	misp-project.org/misp-training/cheatsheet.pdf
Automation & API	REST API and examples	circl.lu/doc/misp/automation
PyMISP	Python library and docs	github.com/MISP/PyMISP
Taxonomies & Galaxies	Vocabularies and clusters	github.com/MISP/misp-taxonomies, .../misp-galaxy
misp-docker	Container operation	github.com/MISP/misp-docker
ZeroMQ	Notification schema	circl.lu/doc/misp/misp-zmq



Questions?

Discussion and Feedback

Langjährige Erfahrung, starke Vernetzung und fundierte MISP-Expertise:



Vernetzung & Zusammenarbeit



Souveräne Threat Intelligence



MISP: Betrieb & Integration

Cyber Threat Intelligence

Gemeinsam souverän und wirksam gestalten

- Langjährige enge Zusammenarbeit mit dem Bundesamt für Sicherheit in der Informationstechnik (BSI)
- Mitglied der Allianz für Cybersicherheit
- DCOM: Herstellerunabhängige Community für den fachlichen Austausch und die gemeinsame Weiterentwicklung im Bereich Cyber Threat Intelligence

Kontakt DCOM: connect@dcso.de

- Herstellerunabhängige Threat Intelligence mit Fokus auf digitale Souveränität
- Fundierte Expertise beim Aufbau und der Weiterentwicklung souveräner TI-Strukturen
- Beratung und Unterstützung rund um MISP

Kontakt TI: connect@dcso.de

- Hosted MISP-Instanzen für einen souveränen und bedarfsgerechten Betrieb
- Integration Engineering zur Anbindung von MISP an bestehende Systeme und Sicherheitsinfrastrukturen
- Unterstützung bei individuellen Integrations- und Automatisierungsvorhaben

Contact

Alexander Heidorn
Head of CERT

+49 151 20 55 24 01
alexander.heidorn@dcso.de

DCSO Deutsche Cyber-Sicherheitsorganisation GmbH
EUREF-Campus 22
10829 Berlin, Germany

TLP Legend

TPL:RED

Personal for named recipients only

In the context of a meeting, for example, RED information is limited to those present at the meeting. The distribution of RED information will generally be via a defined list and in extreme circumstances may only be passed verbally or in person.

TPL:AMBER

Limited distribution

The recipient may share AMBER information with others within their organization and their clients, but only on a 'need-to-know' basis. The originator may be expected to specify the intended limits of that sharing.

TPL:GREEN

Community wide

Information in this category can be circulated widely within a particular community. However, the information may not be published or posted publicly on the Internet, nor released outside of the community. Note: when "community" is not defined, assume the cybersecurity/defense community.

TPL:CLEAR

Unlimited

Subject to standard copyright rules, CLEAR/WHITE information may be distributed freely, without restriction.



TLP:CLEAR